

Domanda **22**

Risposta non
data

Punteggio max.:
7,00

Consider the set of trajectories below, which are obtained by running a given policy in an MDP with three states $S=\{A, B, C\}$ (C is terminal) and two actions $A=\{\text{up}, \text{down}\}$.

A trajectory is a sequence of (S_t, A_t, R_t) :

$(A, \text{up}, 2) \rightarrow (A, \text{down}, 4) \rightarrow (B, \text{down}, 4) \rightarrow (C)$

$(B, \text{up}, 2) \rightarrow (A, \text{down}, 4) \rightarrow (C)$

1. Compute the value functions $V(A)$ and $Q(A, \text{down})$ by resorting to first-visit Monte-Carlo evaluation.
2. Compute the value function $V(A)$ by resorting to Temporal-Difference evaluation. Assume to start from zero value for each state, $\alpha=0.5$, $\gamma=1$.
3. Consider the values $V(A)$ obtained with Monte Carlo and Temporal Difference. Can you tell if, given an infinite number of trajectories, the computed values would be different?

1)

$$V(A) = (10 + 4) / 2 = 7$$

$$Q(A, \text{down}) = (8 + 4) / 2 = 6$$

2)

$$V(A) = 0 + 0.5 * (2 + 0 - 0) = 1$$

$$V(A) = 1 + 0.5 * (4 + 0 - 1) = 2.5$$

$$V(A) = 2.5 + 0.5 * (4 + 0 - 2.5) = 3.25$$

3)

Since TD and MC are consistent estimators, they will converge to the same value $V^{\pi}(A)$

Domanda **23**

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Punteggio max.: 7,00

Consider the set of trajectories below, which are obtained by running a given policy in an MDP with three states $S=\{A, B, C\}$ (C is terminal) and two actions $A=\{\text{up}, \text{down}\}$.

A trajectory is a sequence of (S_t, A_t, R_t) :

$(A, \text{down}, 4) \rightarrow (B, \text{up}, 2) \rightarrow (C)$

$(A, \text{up}, 2) \rightarrow (B, \text{down}, 4) \rightarrow (A, \text{down}, 4) \rightarrow (C)$

1. Compute the value functions $V(A)$ and $Q(A, \text{down})$ by resorting to first-visit Monte-Carlo evaluation.
2. Compute the value function $V(A)$ by resorting to Temporal-Difference evaluation. Assume to start from zero value for each state, $\alpha=0.5$, $\gamma=1$.
3. Consider the values $V(A)$ obtained with Monte Carlo and Temporal Difference. Can you tell if, given an infinite number of trajectories, the computed values would be different?

1)

$$V(A) = (6 + 10) / 2 = 8$$

$$Q(A, \text{down}) = (6 + 4) / 2 = 5$$

2)

$$V(A) = 0 + 0.5 * (4 + 0 - 0) = 2$$

$$V(B) = 0 + 0.5 * (2 + 0 - 0) = 1$$

$$V(A) = 2 + 0.5 * (2 + 1 - 2) = 2.5$$

$$V(A) = 2.5 + 0.5 * (4 + 0 - 2.5) = 3.25$$

3)

Since TD and MC are consistent estimators, they will converge to the same value $V^{\pi}(A)$

Domanda **24**

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Punteggio max.:
7,00

Consider the set of trajectories below, which are obtained by running a given policy in an MDP with three states $S=\{A, B, C\}$ (C is terminal) and two actions $A=\{\text{up}, \text{down}\}$.

A trajectory is a sequence of (S_t, A_t, R_t) :

$(A, \text{up}, 2) \rightarrow (A, \text{down}, 8) \rightarrow (B, \text{down}, 4) \rightarrow (C)$

$(B, \text{up}, 2) \rightarrow (A, \text{down}, 4) \rightarrow (C)$

1. Compute the value functions $V(A)$ and $Q(A, \text{down})$ by resorting to first-visit Monte-Carlo evaluation.
2. Compute the value function $V(A)$ by resorting to Temporal-Difference evaluation. Assume to start from zero value for each state, $\alpha=0.5$, $\gamma=1$.
3. Consider the values $V(A)$ obtained with Monte Carlo and Temporal Difference. Can you tell if, given an infinite number of trajectories, the computed values would be different?

1)

$$V(A) = (14 + 4) / 2 = 9$$

$$Q(A, \text{down}) = (12 + 4) / 2 = 8$$

2)

$$V(A) = 0 + 0.5 * (2 + 0 - 0) = 1$$

$$V(A) = 1 + 0.5 * (8 + 0 - 1) = 4.5$$

$$V(A) = 4.5 + 0.5 * (4 + 0 - 4.5) = 4.25$$

3)

Since TD and MC are consistent estimators, they will converge to the same value $V^{\pi}(A)$

Domanda **25**

Risposta non
data

Punteggio max.:
7,00

Consider the set of trajectories below, which are obtained by running a given policy in an MDP with three states $S=\{A, B, C\}$ (C is terminal) and two actions $A=\{\text{up}, \text{down}\}$.

A trajectory is a sequence of (S_t, A_t, R_t) :

$(A, \text{down}, 4) \rightarrow (B, \text{up}, 2) \rightarrow (C)$

$(A, \text{up}, 2) \rightarrow (B, \text{down}, 4) \rightarrow (A, \text{down}, 8) \rightarrow (C)$

1. Compute the value functions $V(A)$ and $Q(A, \text{down})$ by resorting to first-visit Monte-Carlo evaluation.
2. Compute the value function $V(A)$ by resorting to Temporal-Difference evaluation. Assume to start from zero value for each state, $\alpha=0.5$, $\gamma=1$.
3. Consider the values $V(A)$ obtained with Monte Carlo and Temporal Difference. Can you tell if, given an infinite number of trajectories, the computed values would be different?

1)

$$V(A) = (6 + 14) / 2 = 10$$

$$Q(A, \text{down}) = (6 + 8) / 2 = 7$$

2)

$$V(A) = 0 + 0.5 * (4 + 0 - 0) = 2$$

$$V(B) = 0 + 0.5 * (2 + 0 - 0) = 1$$

$$V(A) = 2 + 0.5 * (2 + 1 - 2) = 2.5$$

$$V(A) = 2.5 + 0.5 * (8 + 0 - 2.5) = 5.25$$

3)

Since TD and MC are consistent estimators, they will converge to the same value $V^{\pi}(A)$

Domanda **26**

Risposta non
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Punteggio max.:
7,00

Consider a linear, hard-margin, two-class SVM classifier defined by parameters $w = [1, 2]$, $b = -2$.
Answer the following questions providing adequate motivations.

- 1) Provide the analytical formula of the boundary and the margins.
- 2) How the point $x_1 = [1; 3/4]$ is classified according to the trained SVM?
- 3) Assume to collect a new sample $x_2 = [1; 3]$ in the negative class. Do you need to retrain the SVM? In which case would you resort to a soft-margin version?

1)

$$x + 2 \cdot x_2 - 2 = 0 \text{ (boundary)}$$

$$x + 2 \cdot x_2 - 3 = 0 \text{ (positive margin)}$$

$$x + 2 \cdot x_2 - 1 = 0 \text{ (negative margin)}$$

2)

$1/2 \rightarrow$ positive class

$$1 + 3/2 - 2 = -0.5 \rightarrow \text{negative class}$$

3)

$1 + 6 - 2 = +5 \rightarrow$ it would be misclassified, thus you need to retrain and x_2 is a SV.
You would use a soft-margin if the problem is no more linearly separable.

Domanda **27**

Risposta non
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Punteggio max.:
7,00

Consider a linear, hard-margin, two-class SVM classifier defined by parameters $w = [-1, 2]$, $b = 2$.

Answer the following questions providing adequate motivations.

- 1) Provide the analytical formula of the boundary and the margins.
- 2) How the point $x_1 = [1; -3/4]$ is classified according to the trained SVM?
- 3) Assume to collect a new sample $x_2 = [1; -3]$ in the positive class. Do you need to retrain the SVM? In which case would you resort to a soft-margin version?

1)

- $x + 2x_2 + 2 = 0$ (boundary)
- $x + 2x_2 + 1 = 0$ (positive margin)
- $x + 2x_2 + 3 = 0$ (negative margin)

2)

- $1 - 3/2 + 2 = -0.5 \rightarrow$ ~~positive~~ **negative** class

3)

- $1 - 6 + 2 = -5 \rightarrow$ it would be misclassified, thus you need to retrain and x_2 is a SV. You would use a soft-margin if the problem is no more linearly separable.

Domanda **28**

Risposta non
data

Punteggio max.:
7,00

Consider a linear, hard-margin, two-class SVM classifier defined by parameters $w = [-1, 2]$, $b = -2$.
Answer the following questions providing adequate motivations.

- 1) Provide the analytical formula of the boundary and the margins.
- 2) How the point $x_1 = [-1; 3/4]$ is classified according to the trained SVM?
- 3) Assume to collect a new sample $x_2 = [-1; 3]$ in the positive class. Do you need to retrain the SVM? In which case would you resort to a soft-margin version?

- 1)
 $-x + 2x_2 - 2 = 0$ (boundary)
 $-x + 2x_2 - 3 = 0$ (positive margin)
 $-x + 2x_2 - 1 = 0$ (negative margin)

- 2)
 $+1 + 3/2 - 2 = 0.5 \rightarrow$ positive class

- 3)
 $1 + 6 - 2 = 5 \rightarrow$ it would be correctly classified, thus you do not need to retrain.
You do not need a soft-margin since the problem is linearly separable.

Domanda **29**

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Punteggio max.: 7,00

Consider a linear, hard-margin, two-class SVM classifier defined by parameters $w = [1, 2]$, $b = 2$. Answer the following questions providing adequate motivations.

- 1) Provide the analytical formula of the boundary and the margins.
- 2) How the point $x_1 = [-1; -3/4]$ is classified according to the trained SVM?
- 3) Assume to collect a new sample $x_2 = [-1; -3]$ in the positive class. Do you need to retrain the SVM? In which case would you resort to a soft-margin version?

1)

$$x + 2x_2 + 2 = 0 \text{ (boundary)}$$

$$x + 2x_2 + 3 = 0 \text{ (positive margin)}$$

$$x + 2x_2 + 1 = 0 \text{ (negative margin)}$$

2)

$$-1 - 3/2 + 2 = -0.5 \rightarrow \text{negative}$$

3)

$-1 - 6 + 2 = -5 \rightarrow$ it would be misclassified, thus you need to retrain and x_2 is a SV. You would use a soft-margin if the problem is no more linearly separable.

Domanda **30**

Risposta non data

Non valutata

Submit your test.

Scegli un'alternativa:

- ☐ a. I want my test to be graded
- ☐ b. I want to withdraw from the test